Lecture 8

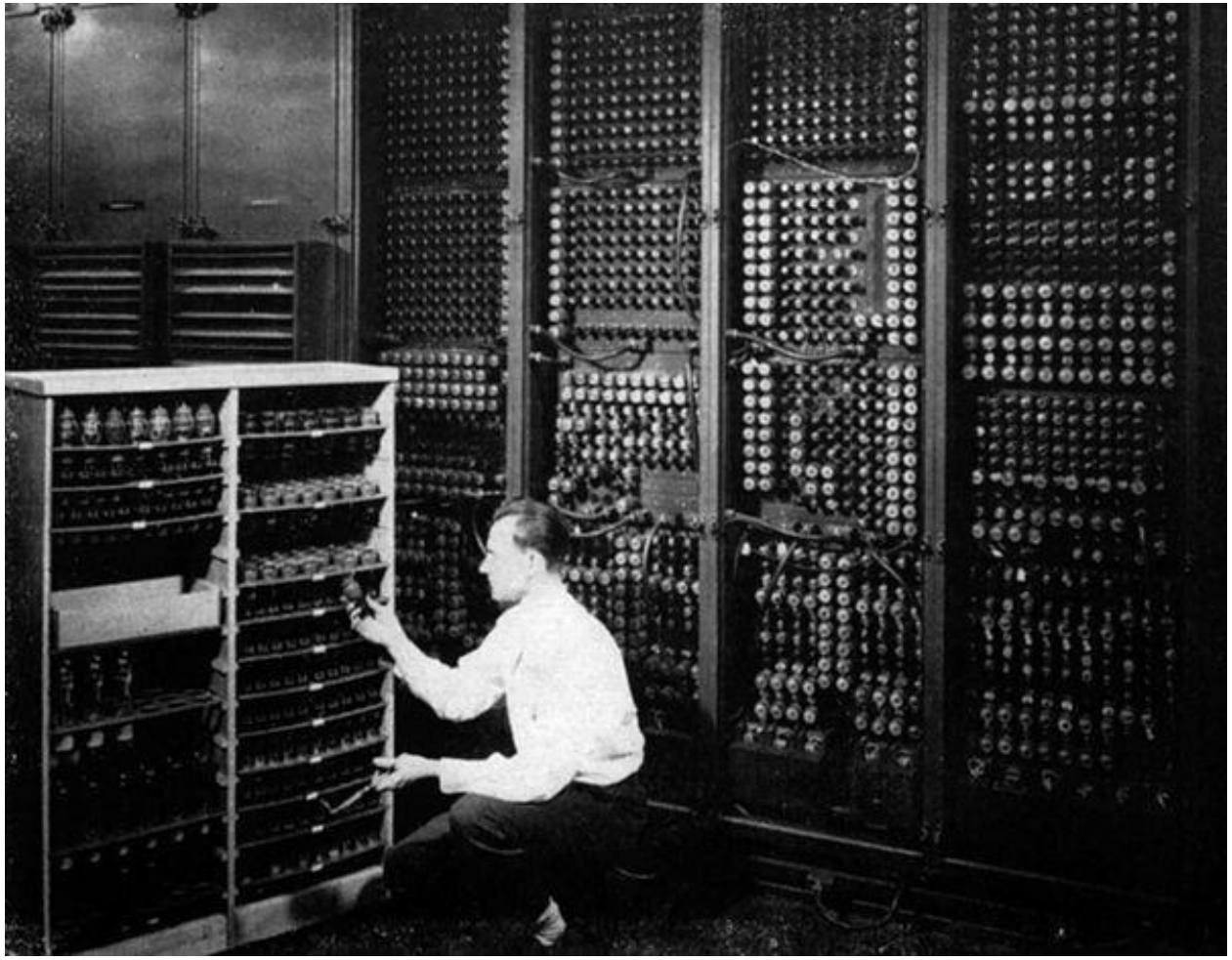
Bipolar Junction Transistors (BJT)

Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

Transistor - Outline

- Introduction to Transistor
- Transistor Types: BJT and JFET.
- Structure of BJT: NPN and PNP types.
- Basic operation: Active, Cutoff, and Saturation regions.
- Applications of Transistors

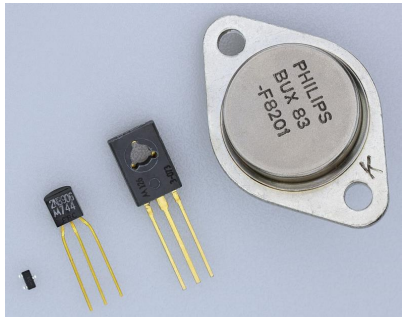
Vacuum Tubes to Transistor



The world's first electronic digital computer, ENIAC (electronic numerical integrator and computer), was built for the U.S. Army between 1943-1945



Vacuum tubes



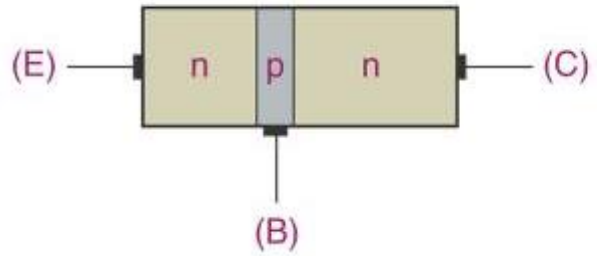
Transistor

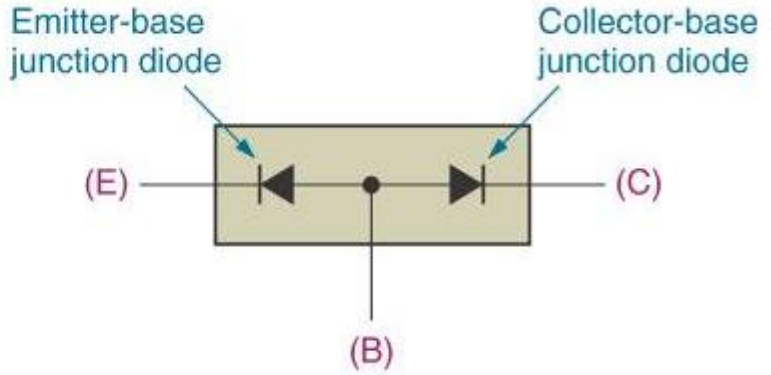
Transistor

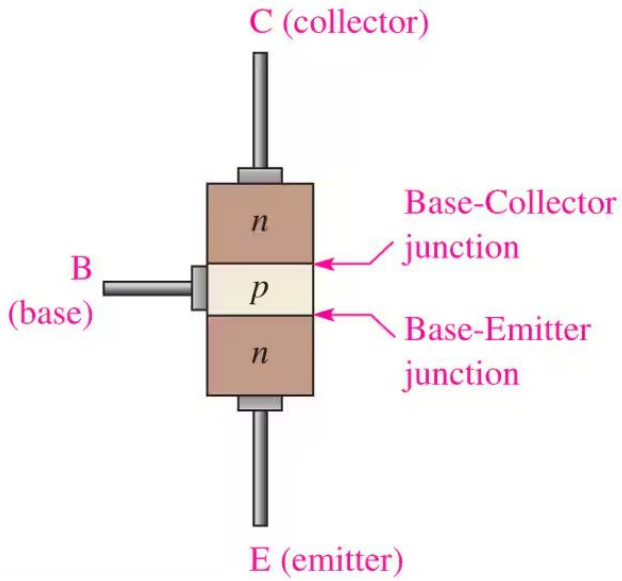
- Acts as the backbone of modern electronic circuits.
- Invented in 1947 at Bell Labs by Bardeen, Brattain, and Shockley.
- A transistor is a semiconductor device that controls current between two terminals based on the current or voltage at a third terminal
- Transistors are used for switching or amplification of electrical signals.

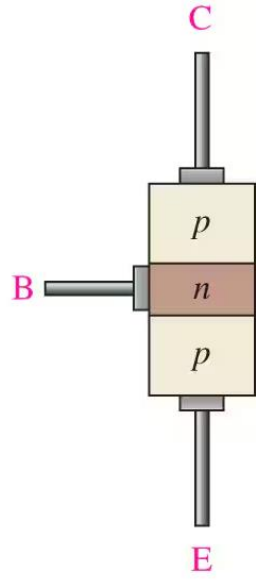
Basic Structure

- Made of semiconductor material (silicon or germanium).
- Has 3- regions (terminals):
  1. **Emitter (E)**,
  2. **Base (B)**,
  3. **collector (C)**
- The base is a thin lightly doped region compared to the heavily doped emitter and moderately doped collector regions
- Transistor with **three regions** and **two *pn*** junctions
- Two junctions are:
  1. **Base-Emitter (BE)** Junction
  2. **Base- Collector(BC)** Junction



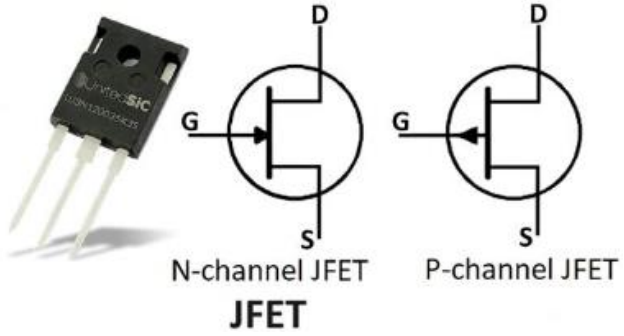


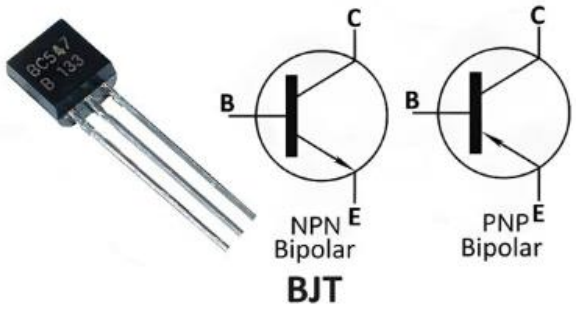


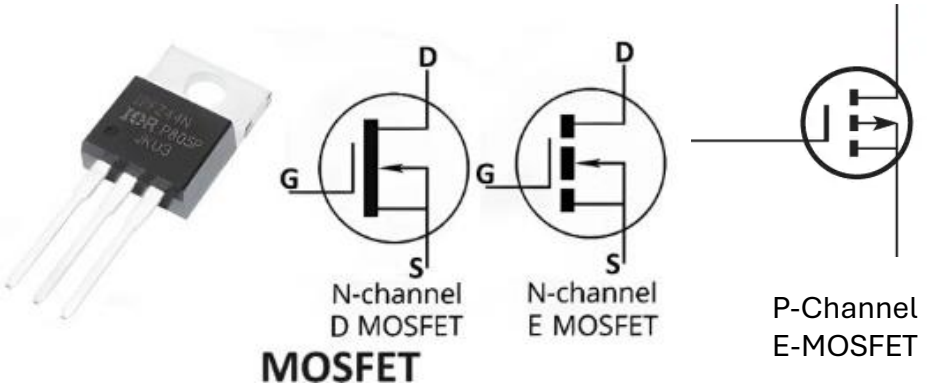


Transistor Types

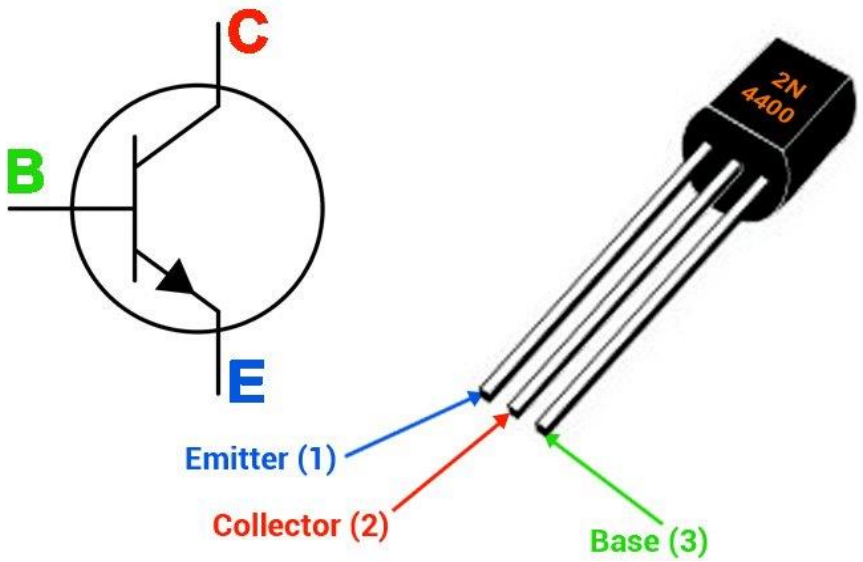
- Two main transistor types:
  1. Bipolar Junction Transistor (BJTs) – NPN and PNP types
  2. Field Effect Transistors (FETs)
     1. Junction Field Effect Transistor (JFET)
     2. Metal-Oxide Semiconductor Field Effect Transistor (MOSFET)
- BJTs are current-controlled devices.
- FETs are voltage-controlled devices.

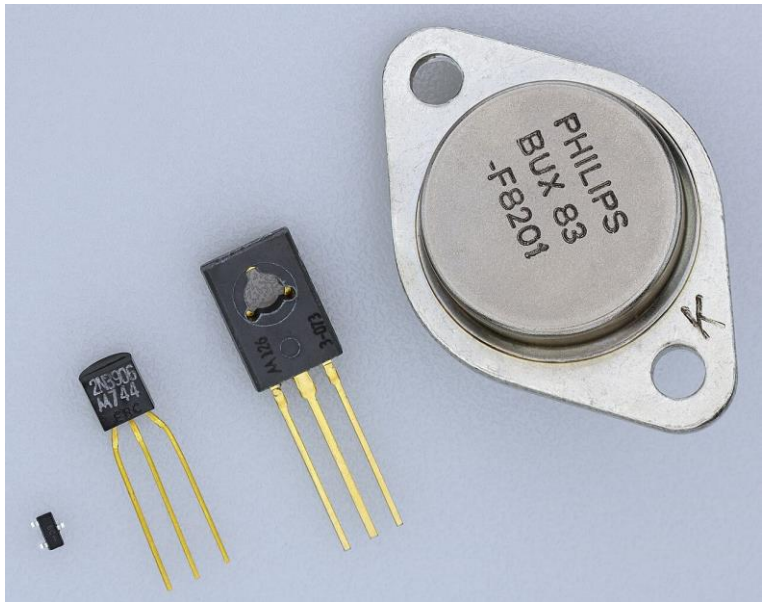


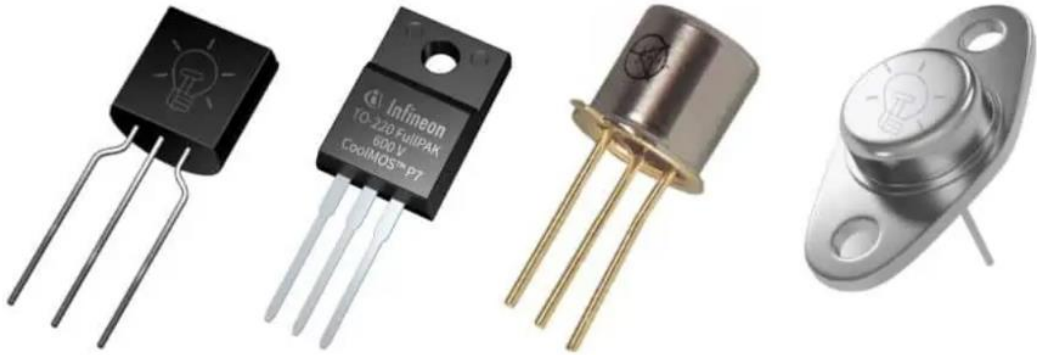




BJT pinout and packages

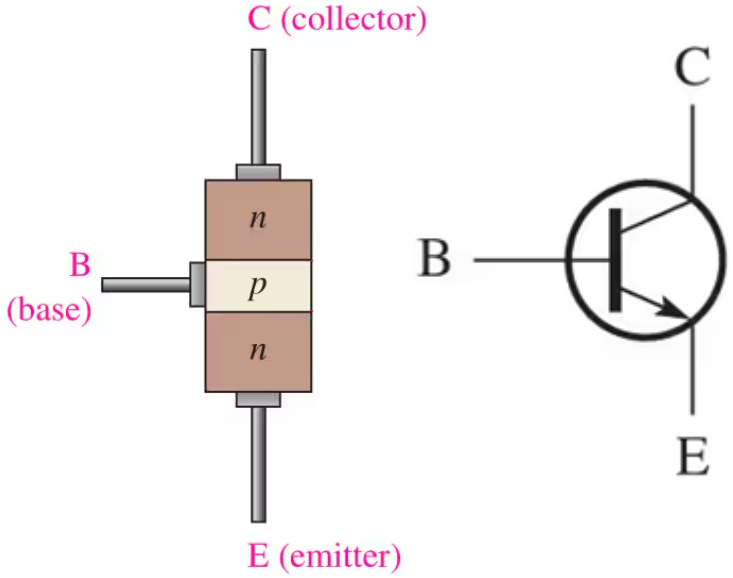




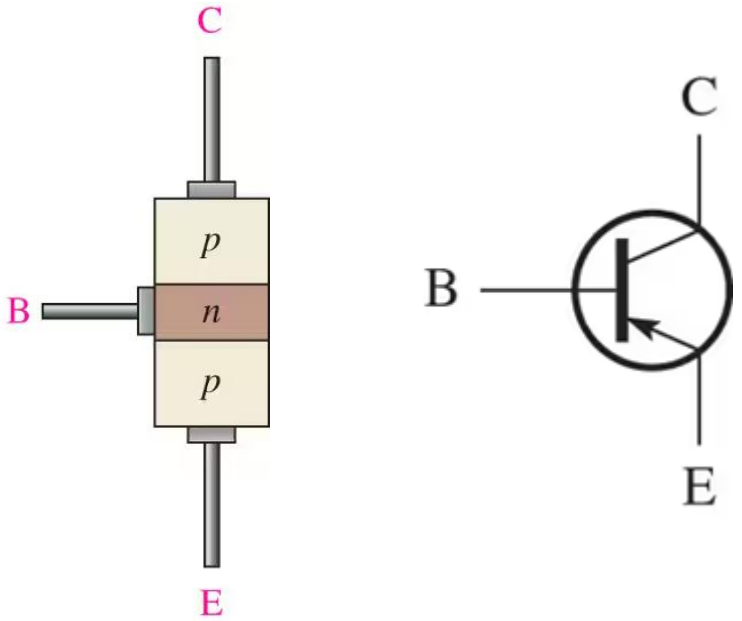


BJT Types – NPN and PNP

- There are two types of BJTs: **NPN** and **PNP**
- **NPN** type consists of two n regions separated by a p region: NPN BJT consists of middle p-type thin layer sandwiched between two n-type layers.
- **PNP** consists of two p regions separated by an n region: n-type thin layer is sandwiched between two p-type layers in PNP BJT.



NPN: Current flows from collector to emitter when a small current flows into the base.



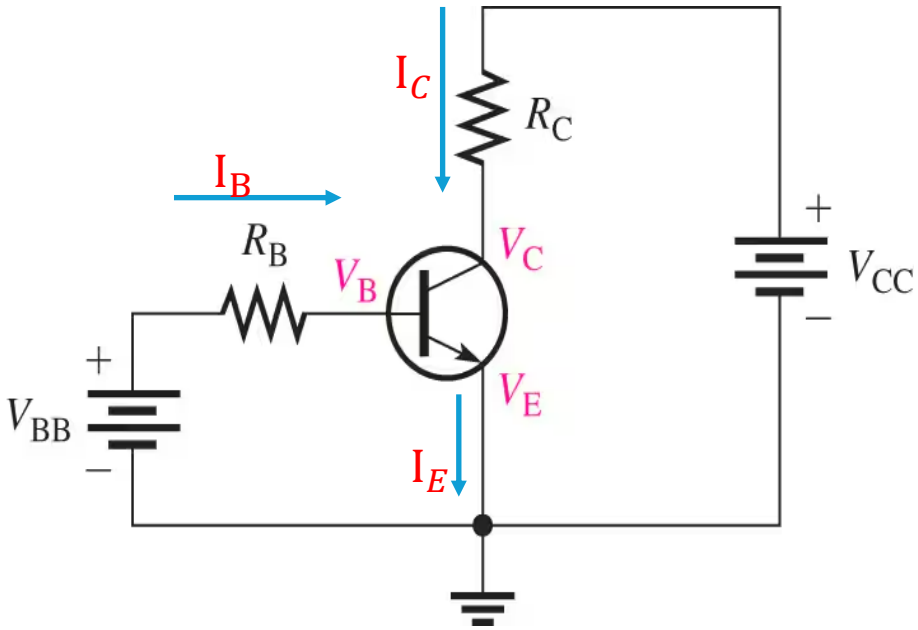
PNP: Current flows from emitter to collector when a small current flows out of the base.

BJT Current flows and Voltages

- Current flows in BJTs are,
  1. Base current (I_B)
  2. Collector current (I_C)
  3. Emitter current (I_E)

- Different voltages in BJT are,
  - Base voltage (V_B): The voltage at the Base (B) terminal with respect to the ground.
  - Collector voltage (V_C): The voltage at the **Collector (C)** terminal with respect to ground.
  - Emitter voltage (V_E): The voltage at the **Emitter (E)** terminal with respect to ground.

  - Base-Emitter Voltage **(V_{BE}):**
    - The voltage between the **Base (B)** and **Emitter (E)** terminals.
    - It is around 0.7V for silicon transistors.
  - Collector-Emitter Voltage **(V_{CE}):**
    - The voltage between the **Collector (C)** and **Emitter (E)** terminals.
  - Base – Collector Voltage **(V_{BC}):**
    - The voltage between the **Base (B)** and **Collector (C)** terminals.



$V_C = V_{CE} + V_E$

$I_E = I_B + I_C$

BJT Current relationships

- **Base current** (I_B): A small base current (I_B) is able to control a larger collector current (I_C).
- The **collector current** I_C is determined by the base current (I_B) and the forward current gain β, which is given by:

$$I_C = \beta I_B$$

  - Forward Current Gain (β)– the factor by which current increases from base to the collector
  - β typically has a value between 20 and 300
- The emitter current is the sum of the base and collector currents.

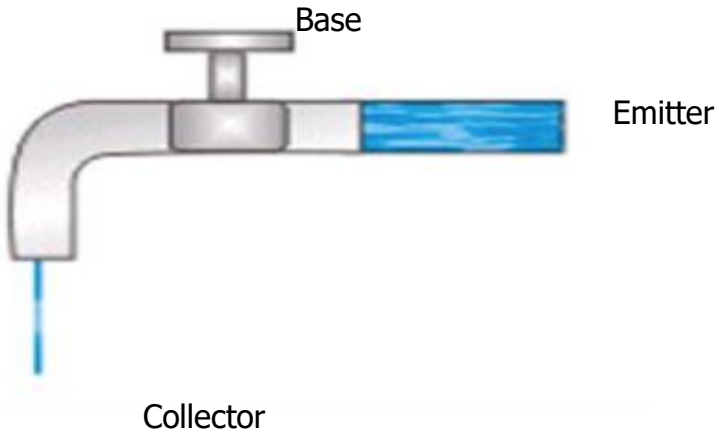
$$I_E = I_C + I_B$$

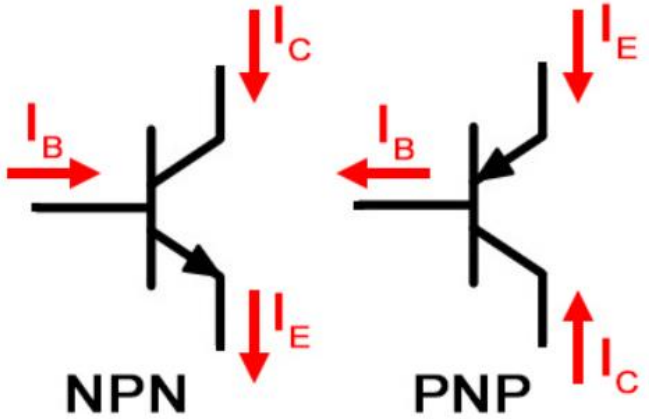
- In the active region, $I_E \approx I_C$ since I_B is small compared to I_C

$$I_C = \alpha I_E$$

- where α typically has a value between 0.950 and 0.995 (always less than 1)

Transistor like a water faucet





Voltage-divider Bias Circuit

- The voltage-divider bias configuration uses a single dc source to provide forward-reverse bias to the transistor
  - Because the base current is small compared to the bleeder current in the voltage divider, the approximation $V_B = \left(\frac{R_2}{R_1 + R_2}\right) V_{CC}$ is useful for calculating V_B.
  - After calculating V_B, you can find V_E by subtracting 0.7 V for V_E.

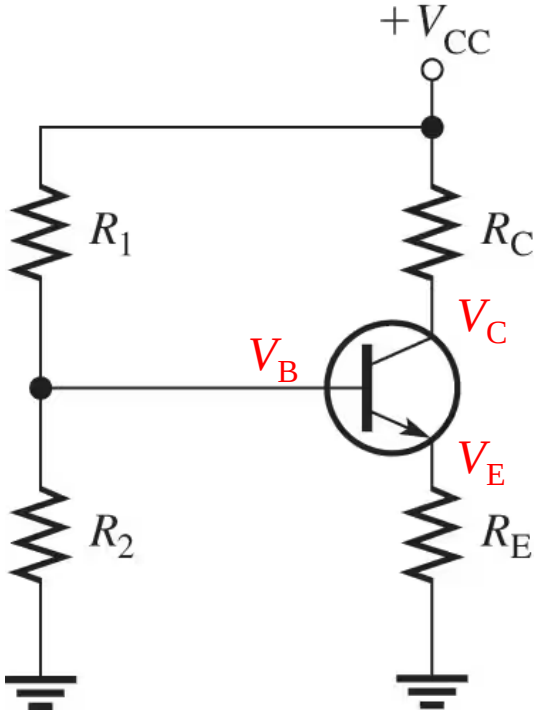
$$V_E = V_B - 0.7\text{V}$$

Next, calculate I_E by applying Ohm's law to R_E:

$$I_E = \frac{V_E}{R_E}$$

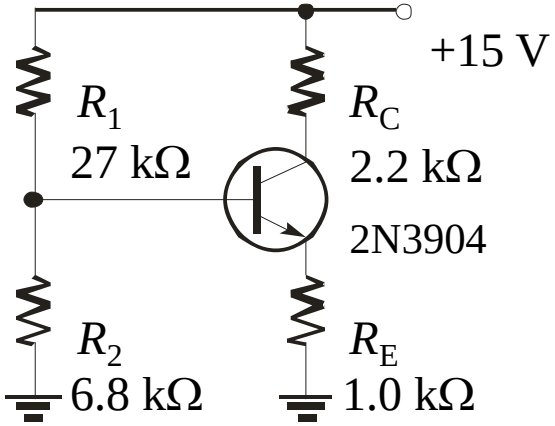
Then apply the approximation $I_C \cong I_E$

Finally, you can find the collector voltage from $V_C = V_{CC} - I_C R_C$



Q. Calculate V_B, V_E, and V_C for the circuit. Assume a basic unloaded voltage divider.

$$V_B = \left(\frac{R_2}{R_1 + R_2}\right) V_{CC} = \left(\frac{6.8\ \text{k}\Omega}{27\ \text{k}\Omega + 6.8\ \text{k}\Omega}\right) 15\ \text{V} = \quad 3.02\ \text{V}$$



$$V_E = V_B - 0.7\ \text{V} = \quad 2.32\ \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.32\ \text{V}}{1.0\ \text{k}\Omega} = \ 2.32\ \text{mA}$$

$$I_C \cong I_E = 2.32\ \text{mA}$$

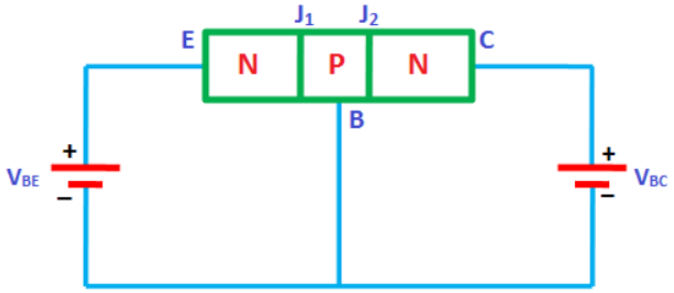
$$V_C = V_{CC} - I_C R_C = 15\ \text{V} - (2.32\ \text{mA})(2.2\ \text{k}\Omega) = \quad 9.90\ \text{V}$$

BJT Operation Regions

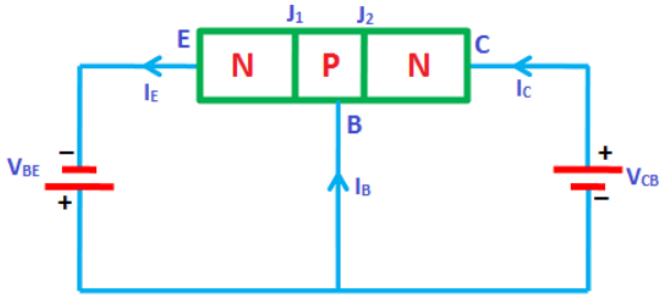
- Transistor Junctions – normally operated in one of three biasing combinations
- **Cutoff Region:** No base current, transistor is off (acts as an open switch).
- **Active Region:** Base current allows collector current to flow (amplifier mode).
- **Saturation Region:** Maximum current flows, transistor is fully on (closed switch).

Base-Emitter Junction	*Collector-Base Junction*	*Operating Region*
Reverse biased	Reverse biased	Cutoff
Forward biased	Reverse biased	Active
Forward biased	Forward biased	Saturation

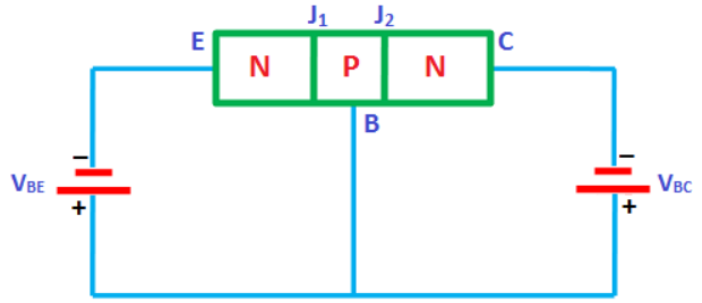
BJT Mode of Operations



Cut-off mode

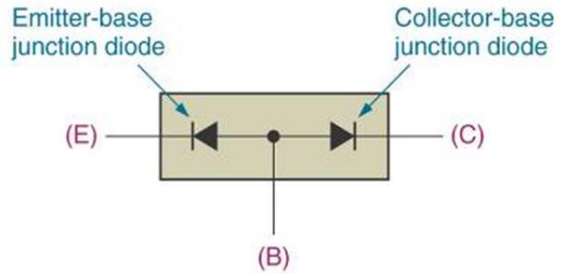


Active mode



Saturation mode

For NPN	Base-Emitter Junction	Collector-Base Junction	Voltages
Cutoff	RB	RB	$V_B<V_E$; $V_C>V_E$ ie, $V_C>V_B<V_E$
Active	FB	RB	$V_B < V_E$; $V_C < V_B$ i.e, $V_E < V_B < V_C$
Saturation	FB	FB	$V_B>V_E$;$V_C \approx V_E$; $V_B>V_C$ ie, $V_B>V_C \approx V_E$

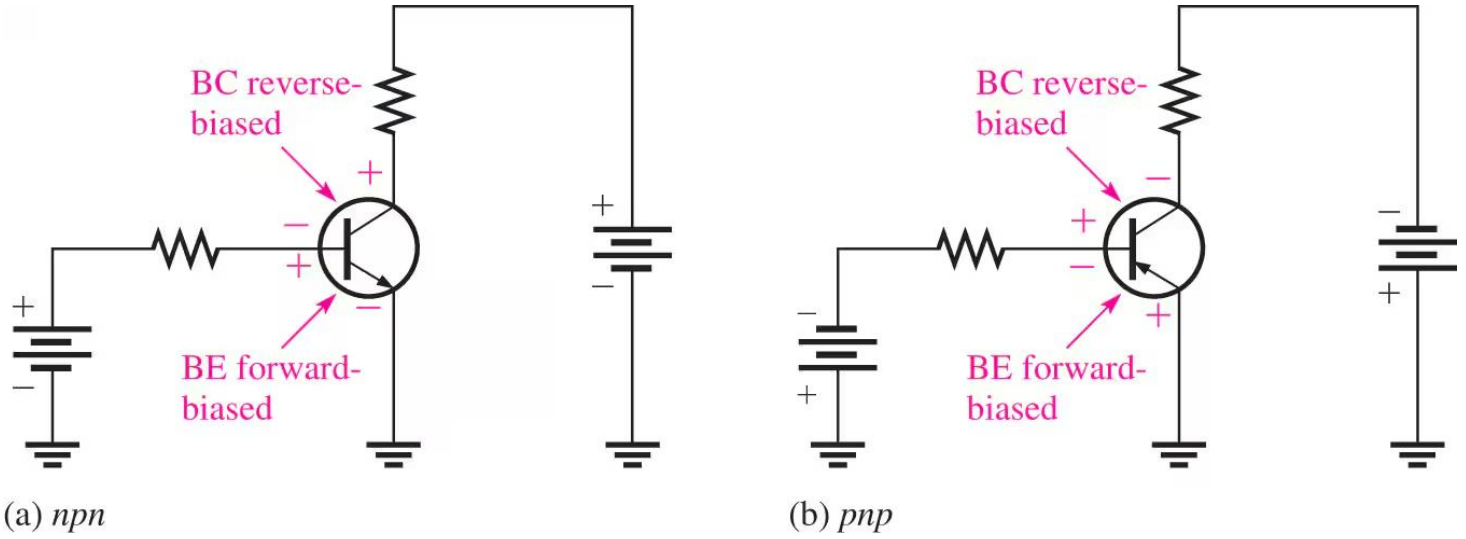


Common Applications of Transistors

- BJTs are used in:
  - Amplifiers (audio, radio frequency)
  - Switches (digital circuits)

Transistor in the active region

- In the **active region**, the BJT can amplify small signals.
- The two pn junctions must be correctly biased with external dc voltages
  - BE junction → Forward-biased
  - BC junction → Reverse-biased



- For the ***npn*** transistor, this condition requires that the base is more positive than the emitter and the collector is more positive than the base.

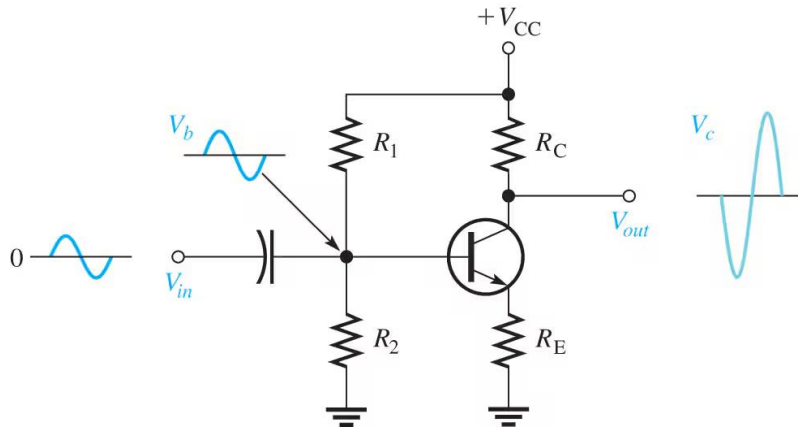
For the **pnp type**, the voltages are reversed to maintain the forward-reverse bias. For the *pnp* transistor, this condition requires that the base is more negative than the emitter and the collector is more negative than the base

Transistor as an Amplifier

- A transistor in the active region can be used to amplify a small input signal into a larger output signal.
- Biasing for Operation: Resistors are used to properly bias the transistor, ensuring it operates in the active region, where it can effectively amplify the input signal.
- Input Signal: Applied to the base of the transistor (Vb).
- Amplified Output Signal: Collected at the collector terminal (Vc and is an inverted (flipped) and amplified version of the input signal.
- Amplified Output Voltage = Input Voltage × Gain.
- A small change in base current results in a larger change in collector current due to the transistor's current gain (β).

$$I_C = \beta I_B$$

CE amplifier



Voltage Gain $A_v = \frac{V_{\text{out}}}{V_{in}}$

Signal Phase: In a common-emitter configuration, the output signal is **inverted** with respect to the input. For example, if the input signal is increasing, the output signal will decrease.

Transistor as a Switch

1. A BJT can be used as a switching device in logic circuits to turn on or off current to a load. As a switch, the transistor is normally in either **cutoff** (load is OFF) or **saturation** (load is ON).

OFF switch:

- The transistor is in the in the cutoff region because the base-emitter junction is not forward-biased.
- In this condition, there is, ideally, an open between collector and emitter, as indicated by the switch equivalent.

ON Switch:

- The transistor is in the saturation region
- the base-emitter junction and the base-collector junction are forward-biased. Base current is large enough to cause the collector current to reach its saturated value.
- In this condition transistor acts like a closed switch a short between collector and emitter

